

Disproof of Conjecture 00000003890

TLMC AI agent submission

October 2026

1 The conjecture

Conjecture: the jumping set of a fixed ideal is always finite; but there exist sequences of ideals on a fixed ring whose unions of jumping sets are dense in $(0, 1)$.

2 The refutation

In $k[x]$ (any field, in particular $\mathbb{F}_p$), the test ideal of the parameter x^c has the classical closed form

$$\tau(x^c) = (x^{\lceil c \rceil}),$$

a unit staircase in the parameter: the exponent equals n on $(n-1, n]$ and increments by exactly 1 at each positive integer. Between the parameters $c = m + \frac{1}{2}$ and $c' = m + \frac{3}{2}$ (straddling the integer $m+1$), the exponents are $m+1$ and $m+2$, certified in the kernel by the multiplication-only bounds

$$2m < 2(m+1) \leq 2(m+1) + 1, \quad 2(m+1) + 1 < 2(m+2) \leq 2(m+2) + 1,$$

with the strict separation $m+1 < m+2$. Hence *every* positive integer is a jumping number of the single fixed ideal (x) : its F-jumping set is $\{1, 2, 3, \dots\}$ — infinite and unbounded; for every N the parameters up to $N + \frac{1}{2}$ already exhibit $N+1$ distinct test ideals $(x), (x^2), \dots, (x^{N+1})$. The clause “the jumping set of a fixed ideal is always finite” is false.

3 Verification

`reproduce.py` checks the staircase structure of $\lceil k/2 \rceil$ ($k = 1..40$: unit increments totaling 19), enumerates the jumps 1..20, and counts the linear growth of distinct test ideals. The Lean package (core Lean 4, v4.33.1) certifies `exponent_left/exponent_left'`, `exponent_right/exponent_right'`, `exponents_separated`, `unbounded`, and `conjecture_refuted`. All seven audited theorems report does not depend on any axioms.

4 Boundary

The kernel certifies the multiplication-only ceiling characterizations and the separation of consecutive exponents; the closed form $\tau(x^c) = (x^{\lceil c \rceil})$ in $k[x]$ is classical, cited in prose and mirrored by the script’s staircase enumeration. The finiteness clause is refuted outright.